
MAE Preliminary Examination

Mathematics Section

Monday, November 13, 2023, 9:00am-11:30noon

Your Name

THREE PROBLEMS WILL BE GRADED	
Select the 3 problems you've worked, to be graded:	points
Problem:	/10
Problem:	/10
Problem:	/10
Total	/30

Please give your answers/work in the space provided

Explain your work/steps clearly

Linear Algebra: Problem 1 [10 points]:

1a) Consider a system of linear time-invariant ordinary differential equations

$$\dot{x}(t) = Sx(t), \text{ where } x(t) \in \mathbb{R}^n$$

for n being a positive integer, and S a skew-symmetric $n \times n$ matrix; that is, $S + S^T = 0$, where T denotes transpose. Show that the trajectory of the system that starts from an initial condition $x(0)$ on the unit sphere, i.e., the Euclidean norm being $\|x(0)\| = 1$, remains on the unit sphere for all times. In other words, you need to prove that $\|x(t)\| = 1$ for all t .

1b) Consider a square matrix $Q \in \mathbb{R}^{n \times n}$ such that for any $x \in \mathbb{R}^n$, the Euclidean norm of Qx is the same as that of x , i.e., $\|Qx\| = \|x\|$. What can you deduce about the matrix Q ? Specifically, you need to specify what is the value of the determinant of Q , what is the inverse of Q , and where are the eigenvalues of Q located on the complex plane.

Workspace for Problem 1: Explain your reasoning/work here.

Solution:

1a) One can argue in several different (equivalent) ways:

Solution i) The solution of the ODE is $x(t) = e^{St}x(0)$, and one can readily see that

$$\begin{aligned} (e^{St})^T e^{St} &= e^{S^T t} e^{St} \\ &= e^{-St} e^{St} = e^{(-S+S)t} \\ &= e^{\text{zero matrix}} = \text{identity matrix.} \end{aligned}$$

Therefore e^{St} is an orthogonal matrix for any t . It follows that

$$\begin{aligned} \|x(t)\|^2 &= (e^{St}x(0))^T e^{St}x(0) \\ &= x(0)^T e^{S^T t} e^{St}x(0) \\ &= x(0)^T x(0) = 1. \end{aligned}$$

Solution ii) We consider the derivative of $\|x(t)\|^2$ and show that this is equal to zero, and therefore, that the length of $x(t)$ remains constant. To this end, we compute

$$\begin{aligned} \frac{d}{dt} x(t)^T x(t) &= \dot{x}(t)^T x(t) + x(t)^T \dot{x}(t) \\ &= x(t)^T S^T x(t) + x(t)^T S x(t) \\ &= x(t)^T (S^T + S) x(t) = 0. \end{aligned}$$

1b) Since $\|Qx\| = \|x\|$ for all x ,

$$x^T Q^T Q x = x^T x,$$

and therefore $Q^T Q = I$, the identity matrix. Therefore, Q is an orthogonal matrix. It can be easily seen that

$$\det(Q^T Q) = (\det Q^T)(\det Q) = 1,$$

and therefore $\det Q = 1$. Also, $Q^{-1} = Q^T$, and lastly, since

$$Qx = \lambda x$$

and x have the same Euclidean length, for any eigenvalue/eigenvector pair, i.e.,

$$\|x\|^2 = (Qx)^* Qx = \overline{(Qx)}^T Qx = |\lambda|^2 \|x\|^2,$$

the eigenvalues must have $|\lambda|^2 = 1$, i.e., they lie on the unit circle of the complex plane.

Workspace for Problem 1: Explain your reasoning/work here.

Solution:

Differential Equations: Problem 2 [10 points]:

Consider the second-order differential equation

$$\ddot{x}(t) + \dot{x}(t) - x(t) + x(t)^2 = 0.$$

- i) Write a corresponding state-space representation in the form of two first order differential equations, with position $x(t)$ and velocity $\dot{x}(t)$ as state variables, as you will consider the dynamics on the phase plane. Having done that, determine:
 - ii) critical points, i.e., points of equilibrium.
 - iii) linearization of the differential equations about each point of equilibrium
 - iv) the eigenvalues and any real eigenvectors of the corresponding linearized models
 - v) the type of critical point each is (e.g., stable, unstable, focus, saddle point).
 - vi) draw as best as you can the phase portrait. It is important to indicate correctly the rotation about foci, i.e., clockwise or counterclockwise.

Solution:

i) with

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} = \begin{bmatrix} x \\ \dot{x} \end{bmatrix}$$

we get the state-space representation

$$\begin{pmatrix} \dot{\mathbf{x}}_1 \\ \dot{\mathbf{x}}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{x}_2 \\ -\mathbf{x}_2 + \mathbf{x}_1 - \mathbf{x}_1^2 \end{pmatrix} =: f(\mathbf{x}).$$

ii) For $f(\mathbf{x}) = 0$ we get that $\mathbf{x}_2 = \dot{x} = 0$ while $\mathbf{x}_1 \in \{0, 1\}$, for two points of equilibrium, respectively.

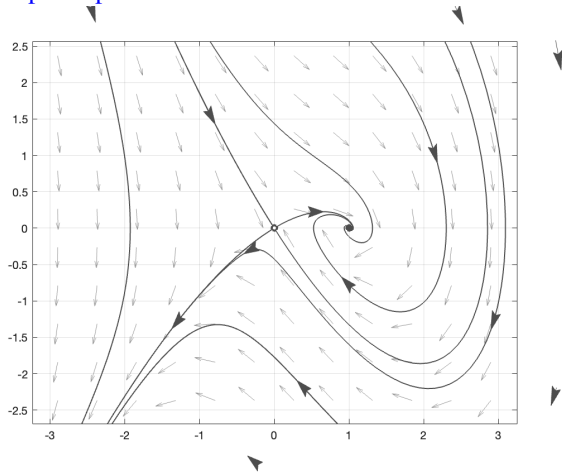
iii-v) The Jacobian is

$$\partial f / \partial \mathbf{x} = \begin{pmatrix} 0 & 1 \\ 1 - 2\mathbf{x}_1 & -1 \end{pmatrix}.$$

Thus, at:

equilibrium point	linearized dynamics	eigenvalues/vectors	type of equilibrium
$\mathbf{x}_{\text{eq. } 1} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$:	$\dot{\mathbf{x}} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} \mathbf{x}$	$\lambda_{1,2} = \frac{-1 \pm \sqrt{5}}{2}, v_{1,2} \in \left\{ \begin{pmatrix} -0.52 \\ 0.85 \end{pmatrix}, \begin{pmatrix} 0.85 \\ 0.52 \end{pmatrix} \right\}$	saddle
$\mathbf{x}_{\text{eq. } 2} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$:	$\dot{\mathbf{x}} = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} \mathbf{x}$	$\lambda_{1,2} = \frac{-1 \pm i\sqrt{3}}{2}, v_{1,2} \text{ complex}$	stable focus

v) The phase portrait follows:



To verify clockwise rotation about the stable focus, observe that to the right of the particular point of equilibrium, i.e., for $\dot{x} = 0$ and $x > 1$, it holds that $\ddot{x} < 0$, i.e., $\dot{\mathbf{x}}_2 < 0$ and the vector field points towards decreasing values of $\mathbf{x}_2$.

Workspace for Problem 2: Explain your reasoning/work here.

Solution:

Problem 3 [10 points]:

Consider the wave equation

$$u_{tt} - a^2 \Delta u = 0$$

in the cubic domain $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$, where a is a positive constant.

The boundary conditions are

$$\frac{\partial u}{\partial n} = 0 \quad \text{on five sides of the cube at } x = 0, x = 1, y = 0, y = 1, \text{ and } z = 0$$

and

$$u = 0 \quad \text{on } z = 1$$

The initial conditions are

$$u(0, x, y, z) = (1 - 2y)^2 \cos(\pi z/2)$$

and

$$u_t(0, x, y, z) = (1 - 2z)^2$$

Using separation of variables and eigenfunction expansion, find $u(t, x, y, z)$. The coefficients of series expansions can be left in integral form. You may notice that the initial and boundary conditions are uniform in the x direction.

Workspace for Problem 3: Explain your reasoning/work here.

Solution:

$$\frac{\partial^2 u}{\partial t^2} = a^2 \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right]$$

Assume

$$u = \sum \sum \sum u_{lmn}$$

$$u_{lmn} = X(x)Y(y)Z(z)T(t)$$

For linear equation

$$\frac{\partial^2 u_{lmn}}{\partial t^2} = a^2 \nabla^2 u_{lmn}$$

Differentiate and divide by u_{lmn}

$$\frac{T''}{T} = a^2 \left(\frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} \right)$$

$$Z_n'' + \nu_n^2 Z_n = 0; \quad Z_n'(0) = 0, \quad Z_n(1) = 0$$

$$Z_n = \cos \nu_n z$$

$$\nu_n = (2n + 1) \frac{\pi}{2}, \quad n = 0, 1, 2, \dots$$

$$Y_m'' + \mu_m^2 Y_m = 0; \quad Y_m'(0) = 0, \quad Y_m'(1) = 0$$

$$Y_m = \cos \mu_m z$$

$$\mu_m = m\pi, \quad m = 0, 1, 2, \dots$$

$$X_l'' + \lambda_l^2 X_l = 0; \quad X_l'(0) = 0, \quad X_l'(1) = 0$$

$$X_l = \cos \lambda_l z$$

$$\lambda_l = l\pi, \quad l = 0, 1, 2, \dots$$

$$T'' + a^2(\lambda_l^2 + \mu_m^2 + \nu_n^2)T = 0$$

$$T = \sin \omega_{lmn} t, \cos \omega_{lmn} t$$

where

$$\omega_{lmn} = a\sqrt{\lambda_l^2 + \mu_m^2 + \nu_n^2} = a\pi\sqrt{l^2 + m^2 + [(2n+1)/2]^2}$$

By the principle of superposition, we have

$$u = \sum_{l=0}^{+\infty} \sum_{m=0}^{+\infty} \sum_{n=0}^{+\infty} (A_{lmn} \cos \omega_{lmn} t + B_{lmn} \sin \omega_{lmn} t) \cos \lambda_l x \cos \mu_m y \cos \nu_n z$$

Using the initial conditions, we get

$$u(0, x, y, z) = \sum_{l=0}^{+\infty} \sum_{m=0}^{+\infty} \sum_{n=0}^{+\infty} A_{lmn} \cos \lambda_l x \cos \mu_m y \cos \nu_n z = (1-2y)^2 \cos(\pi z/2)$$

$$\frac{\partial u}{\partial t}(0, x, y, z) = \sum_{l=0}^{+\infty} \sum_{m=0}^{+\infty} \sum_{n=0}^{+\infty} \omega_{lmn} B_{lmn} \cos \lambda_l x \cos \mu_m y \cos \nu_n z = (1-2z)^2$$

$A_{lmn} = 0$ for all l and n except $l = 0$ and $n = 0$.

$B_{lmn} = 0$ for all l and m except $l = 0$ and $m = 0$.

Denoting

$$A_m = A_{0m0}$$

$$B_n = B_{00n}$$

$$\omega_m = \omega_{0m0} = a\pi\sqrt{m^2 + (1/2)^2}$$

and

$$\omega_n = \omega_{00n} = a\pi\frac{2n+1}{2}$$

we have

$$\begin{aligned} u(t, x, y, z) &= \cos(\pi z/2) \sum_{m=0}^{+\infty} A_m \cos \omega_m t \cos \mu_m y + \sum_{n=0}^{+\infty} B_n \sin \omega_n t \cos \nu_n z \\ &= \cos(\pi z/2) \sum_{m=0}^{+\infty} A_m \cos(a\pi\sqrt{m^2 + (1/2)^2} t) \cos(m\pi y) + \sum_{n=0}^{+\infty} B_n \sin[a(2n+1)\frac{\pi}{2} t] \cos[(2n+1)\frac{\pi}{2} z] \end{aligned}$$

where

$$A_0 = \int_0^1 (1-2y)^2 dy$$

$$A_m = 2 \int_0^1 (1-2y)^2 \cos \mu_m y dy, \quad m = 1, 2, \dots,$$

$$B_n = \frac{2}{\omega_n} \int_0^1 (1-2z)^2 \cos \nu_n z dz \quad n = 1, 2, \dots,$$

Essentially, we have two solutions, one from the two-dimensional initial displacement condition

$u(t=0) = f(y) \cos(\pi z/2)$, and the other for the one-dimensional initial velocity condition $\frac{\partial u}{\partial t}(t=0) = g(z)$

Alternatively, one may simplify the above solution procedure if we start by solving a 2-dimensional problem in the y - z plane on recognizing the independence of the problem from the x coordinate. The final solution will, of course, be the same.

Workspace for Problem 3: Explain your reasoning/work here.

Solution:

Problem 4 [10 points]:

We have the heat equation

$$u_t - \alpha u_{xx} = 0 \quad \text{in the finite domain } 0 \leq x \leq 1 \text{ and } t \geq 0$$

with the initial condition: $u(0, x) = 0$ and the boundary conditions: $u(t, 0) = h_0$ and $u(t, 1) = 0$, where h_0 is a constant. Solve for $u(t, x)$ using Laplace transform. Define any integral function that use.

You may find the following series and Laplace Transform Table useful.

$$\frac{1}{1 - \epsilon} = \sum_{n=0}^{+\infty} \epsilon^n, \quad \text{for } |\epsilon| < 1$$

Table 1: Some Useful Laplace Transforms

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
1	$\frac{1}{s}$
$t^n, \quad n = 0, 1, 2, \dots$	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a} \quad s > a$
$\sin at$	$\frac{a}{s^2+a^2} \quad s > 0$
$\cos at$	$\frac{s}{s^2+a^2} \quad s > 0$
$H(t-a)$	$\frac{e^{-as}}{s} \quad s > 0$
$H(t-a)f(t-a)$	$e^{-as}F(s)$
$e^{at}f(t)$	$F(s-a)$
$f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)d\tau$	$F(s)G(s)$
$f^n(t)$	$s^n F(s) - s^{n-1}f(0) - \dots - f^{n-1}(0)$
$\int_0^t f(\tau)d\tau$	$\frac{1}{s}F(s)$
$\text{erf}(t/2a)$	$\frac{1}{s}e^{a^2s^2} \text{erfc}(as)$
$\text{erfc}(\frac{a}{2\sqrt{t}})$	$\frac{1}{s}e^{-a\sqrt{s}}$
$\frac{1}{\sqrt{\pi t}}$	$1/\sqrt{s}$
$\frac{1}{\sqrt{\pi t}}e^{-\frac{a^2}{4t}}$	$e^{-a/\sqrt{s}}/\sqrt{s}$
$t^{-1/2}e^{-a^2/4t}$	$\sqrt{\frac{\pi}{s}}e^{-a\sqrt{s}} \quad a \geq 0$
$t^{-3/2}e^{-a^2/4t}$	$\frac{2\sqrt{\pi}}{a}e^{-a\sqrt{s}} \quad a > 0$
$2\sqrt{\frac{t}{\pi}} \exp\left(-\frac{a^2}{4t}\right) - a \text{erfc} \frac{a}{2\sqrt{t}}$	$\frac{1}{s^{3/2}}e^{-a\sqrt{s}} \quad a \geq 0$

Workspace for Problem 4: Explain your reasoning/work here.

Solution:

Performing the Laplace transform on the PDE and the boundary conditions we get

$$sU - \alpha U_{xx} = 0$$

where $U(x, s) = \mathcal{L}u$ with the boundary conditions

$$U(0, s) = h_0/s$$

$$U(1, s) = 0$$

The solutions is

$$U(x, s) = \frac{h_0}{s} \frac{e^{-\sqrt{\frac{s}{\alpha}}x} - e^{-\sqrt{\frac{s}{\alpha}}(2-x)}}{1 - e^{-2\sqrt{\frac{s}{\alpha}}}}$$

$$U(x, s) = \frac{h_0}{s} \left(e^{-\frac{x}{\sqrt{\alpha}}\sqrt{s}} - e^{-\frac{2-x}{\sqrt{\alpha}}\sqrt{s}} \right) \sum_{n=0}^{+\infty} e^{-\frac{2n}{\sqrt{\alpha}}\sqrt{s}} = h_0 \sum_{n=0}^{+\infty} \left(\frac{1}{s} e^{-\frac{x+2n}{\sqrt{\alpha}}\sqrt{s}} - \frac{1}{s} e^{-\frac{2(n+1)-x}{\sqrt{\alpha}}\sqrt{s}} \right)$$

Notice the inverse transform

$$\mathcal{L}^{-1} \left\{ \frac{1}{s} e^{-a\sqrt{s}} \right\} = \operatorname{erfc} \left(\frac{a}{2\sqrt{t}} \right)$$

We get

$$u(x, t) = h_0 \sum_{n=0}^{+\infty} \left[\operatorname{erfc} \left(\frac{x+2n}{\sqrt{4\alpha t}} \right) - \operatorname{erfc} \left(\frac{2(n+1)-x}{\sqrt{4\alpha t}} \right) \right]$$

where $\operatorname{erfc}(z)$ is the complimentary error function

$$\operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}} \int_z^{+\infty} e^{-\eta^2} d\eta$$

Thus,

$$u(x, t) = \frac{2h_0}{\sqrt{\pi}} \sum_{n=0}^{\infty} \left[\int_{\frac{x+2n}{\sqrt{4\alpha t}}}^{\infty} e^{-\eta^2} d\eta - \int_{\frac{2(n+1)-x}{\sqrt{4\alpha t}}}^{\infty} e^{-\eta^2} d\eta \right] = \frac{2h_0}{\sqrt{\pi}} \sum_{n=0}^{\infty} \int_{\frac{x+2n}{\sqrt{4\alpha t}}}^{\frac{2(n+1)-x}{\sqrt{4\alpha t}}} e^{-\eta^2} d\eta$$

Workspace for Problem 4: Explain your reasoning/work here.

Solution: